

Interview with Mary Tsingou Menzel, November 22, 2024

T. Dauxois^{1*}, A. Menzel, A. Ponno^{2†}, S. Ruffo^{3‡}, and M.
Tsingou-Menzel[§]

¹ CNRS, 75016 Paris, France

² Università di Padova, Dipartimento di Matematica, Via Trieste 63, 35121
Padova, Italy

³ ISC-CNR, via Madonna del Piano 10, 50019 Sesto Fiorentino, Italy

September 3, 2026

Abstract

Mary Tsingou Menzel was interviewed on November 22, 2024, at her Los Alamos home, with the assistance of her daughter, Ann Menzel.

Introduction

The Fermi-Pasta-Ulam (FPU) problem, first simulated on an early electronic computer between 1953 and 1955, heralded the dawn of both non-linear physics and nonequilibrium statistical physics, laying the groundwork for decades of future discovery. The *numerical experiment* investigated how energy distributes across a chain of atoms in the presence of weak nonlinearities. Surprisingly, the system did not evolve toward energy equipartition as

*email: thierry.dauxois@cnrs.fr

†email ponno@math.unipd.it

‡email ruffo@sissa.it

§Deceased on August 27, 2026.

expected, but instead periodically returned to its initial state - a phenomenon now known as the FPU paradox.

This experiment opened two major research paths:

- the theory of *solitons* (nonlinear localized waves);
- the study of *chaos* and *quasi-integrable Hamiltonian systems*.



Figure 1: Mary Tsingou Menzel in 2019.

At the heart of this first-ever numerical experiment was Mary Tsingou (see Figure 1), the mathematician and programmer who developed and wrote the code on the MANIAC I computer at Los Alamos.

As previously noted in [1], although she performed the primary work for the original report [2], her name appeared only as a “co-worker” in a footnote rather than as a formal author (see Figure 3). In the reprinted version of the report, among the Collected papers of Enrico Fermi [3], her name was removed from the list of authors. Interestingly, we found a previous internal report of Los Alamos, LA-1557, by J. Pasta and S. Ulam [4], where she

does not appear both among the co-workers and among the authors, but her contribution is there mentioned in the text (see Figure 2). It is important to emphasize that Mary Tsingou was later referred to under her married name, M.T. Menzel (for Mary Tsingou-Menzel), which further obscured her scientific identity (see e.g. [5]).

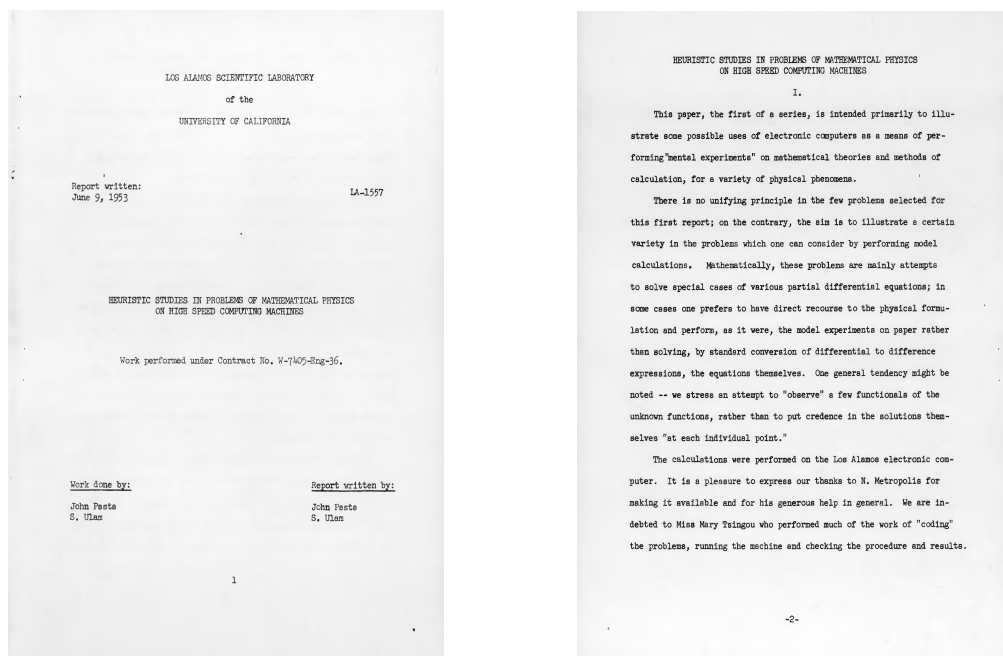


Figure 2: Heading page (left) and page 2 (right) of the original LA-1557 report by Pasta and Ulam [4]. While Mary Tsingou is not listed on the heading page, her contributions are acknowledged in detail on page 2.

Born in 1928 in Milwaukee to a Greek family, she fled Europe just before World War II. After earning degrees in mathematics, she was recruited at Los Alamos National Laboratory in the early 1950s, during the Korean War. There, she quickly established herself as one of the first female programmers, later mastering FORTRAN and contributing to high-impact scientific and defense projects, including the Strategic Defense Initiative - commonly known as the "Star Wars" program - under President Reagan.

Mary Tsingou is now widely recognized as a key contributor to this land-

Acknowledgement

This paper is dedicated to the memory of Mary Tsingou Menzel, who passed away during the completion of this manuscript.

References

- [1] T. Dauxois, Fermi, Pasta, Ulam, and a mysterious lady, *Physics Today* 61 (2008), 55-57.
- [2] E. Fermi, J. Pasta, S. Ulam and M. Tsingou, *Studies in nonlinear problems*, Los Alamos internal report, document LA-1940 (May 1955); available at
<https://www.osti.gov/servlets/purl/4376203>
- [3] E. Fermi, J. Pasta and S. Ulam, *Studies in nonlinear problems*; reprinted in in "Collected papers of Enrico Fermi", ed. by The University of Chicago Press (1965); vol. 2, work n. 266, 977-988.
- [4] J. Pasta and S. Ulam, *Heuristic studies in problems of mathematical physics on high speed computer machines*, Los Alamos internal report, document LA-1557 (June 1953); available at
<https://www.osti.gov/servlets/purl/4412600>
- [5] J.L. Tuck and M.T. Menzel, The superperiod of the nonlinear weighted string (FPU) problem, *Advances in Mathematics* 9 (1972), 399-407.
- [6] "We thank Miss Mary Tsingou", by Virginia Grant.
<https://www.lanl.gov/media/publications/national-security-science/1220-we-thank-miss-mary-tsingou>
- [7] "Oral-History: Mary Tsingou Menzel", by Janet Abbate.
https://ethw.org/Oral-History:Mary_Tsingou_Menzel

The interview

Interviewers : When did you go to Los Alamos, and why?

Mary : *The group that I was in, was doing the calculations, and they were just interested in people that knew a little bit about mathematics. They didn't want anything special. They were going to train us to do some hand calculations.*

I : What year was it?

M : *I came in 1952. In '48, I graduated from high school. So it was '52 when I graduated from college, and that's when I came to Los Alamos. Most of my friends that graduated that year were here by fall because they had their clearances done in three months. I was among the last group of graduates to arrive and my clearance didn't come through till mid-November, just before Thanksgiving. So, I told them it was no use coming, that I was going to come the first of the year. So my co-workers, the women that were working, they had arrived six months earlier than I.*

I : Why did you go to Los Alamos?

M : *Well, they had overtime because they were desperately..., they were in a rush to develop a hydrogen bomb. And so they were working, we were working Saturdays. And so we got overtime on Saturdays. So I thought it was really great to make all that money.*

I : How much were you making an hour when you started?

M : *I don't remember how much I was making, but it was more than General Electric would pay. So I kept General Electric, I kept telling them I wasn't making a decision yet. I kept them on a hook because in case this job fell through, I could go back to New York with General Electric. We were getting overtime Saturdays. So I thought, well, that's great.*

I : Why didn't you buy a car? ¹

M : *At that time, you have to buy stuff. It wasn't like a lease. So my goal was to save money so I could buy a car, so I could get around. I moved into the dorm when I came here.*

¹The question was suggested by Ann Menzel, who recalled how her mother was strongly motivated to buy a car at that time.

I : Who else was living in the dorm with you? Who you worked with?

M : *Oh, Fermi's daughter was there that summer. She stayed there because she didn't want to stay with her parents. See, down the hall, when you go down the street, there was, or at the end, where it's the president's..., Bathtub Row. Well, the big house right in the middle was Stan Ulam's, and I did work for him. He had this guy, John Pasta, who was working with him. Pasta would set up the mathematics part of it, and Stan Ulam would come in only, work a few hours a day and then go home for some reason. They were so glad to have him at the lab, that he could set his own hours.*

I : When you arrived, it was '52, so when did you start working on the Fermi-Pasta-Ulam-Tsingou problem?

M : *Almost immediately, I think, because the machine went out and they themselves said, "we've got the computer now, we've got to think of some problems we can use the computer for." So they came up with this vibrating string problem, and they were surprised in the fact that it didn't disperse. It stayed in the first two modes and remained...it kept going back and forth. Their expectation was that it would spread across, but it just stayed in the first two modes, going back and forth. And that was a big surprise, so...but then they moved on. They weren't...you know, it was just one of the problems.*

I : How old were you during that time?

M : *Well, I was 23, I guess. I think when I came, I was 22, and I had my birthday in October. I graduated from high school at 18. Actually, for a while, they were ignoring the women. They were going to colleges and recruiting the young men who were majoring in mathematics. But the Korean War was on, and so they were being drafted. So they come here for a few months, and by the time they trained them, they'd be called into service, and so they had to leave. That's why they decided to start actually recruiting women rather than men, so that they'd be around for a little longer period, rather than with the men. As soon as I came, for some reason, the Korean War stopped and they stopped recruiting. So the guys that were here stayed here. There were a few young guys that were doing the same thing we were doing.*

I : Can you say a few more words about the daughter of Fermi? How it comes that you got along very well with each other?

M : *They had one of the homes on Bathtub Row. She didn't want to stay with her parents. So she actually moved into the dormitory. So she was with us, young people, rather than with her parents.*

I : She was working in your group?

M : *No, I don't think she worked at all. She was in there, associated with the lab. She was just living in the dorm to be away from her parents. So she was around young people. Well, we worked six days a week. But we planned on going some place around here, one of the sights to see as a group. So they planned it. We'd leave right after Saturday, right after work, and drive up there, wherever we were going, probably in Colorado, and arrive there at midnight and stuff. And we'd leave Saturday night, get there Sunday morning, be there for Sunday, and then we'd have to be back, and then drive back that afternoon, and be back to work on Monday.*

I : What about the trip you took to the nuclear test with all the girls? You went to Las Vegas or something? Do you remember that?

M : *Yeah. As a field trip. Well, what happened was some of the people, the older people, they were older than 30, but they had been here. They were actually flown out to witness the atomic explosion. And so one of the young men was going to drive out. And so actually, there was four of us girls. He was the driver, and four of us girls decided to go and put our names out. And we were treated just like top executives there. They welcomed us and took us around and showed us where the previous explosions were, the data and all that stuff, and then took us to the various locations where they had dropped our weapons. And then they took us way out, about, was it 9, 10 miles away from where the explosion was going to be. We were out in one site. They actually had chairs around to sit down, but we were the only ones out there that day, this young group of people. And it was pretty dark at night, and then all of a sudden they dropped a bomb, and it was almost like daylight. We were about, was it 9 or 13 miles away from where they dropped the bomb. And it lit up the whole area, because it was dark. It was early in the morning. It was completely dark, but when they dropped it, it just lit up the whole area. And then a few minutes later, the air came, and it sort of pushed me. I sat down onto the bench, but I didn't realize until it was a few seconds. It must have been just seconds, but it looked like it was much later that I felt the pressure that pushed me, and I landed on my bench there.*

I : It was an edge bomb?

M : *They had people from the armed forces out there, and they had brought some temporary housing to see how it would stand. I don't know why they thought it would stand. It was completely demolished. There was nothing*

there. It showed that supposedly one of these portable housing units were out there. And they had part of the army there undercover, and after the bomb dropped, they were supposed to get up and be ready to fight, I guess, be ready for battle in case they were dropped someplace. They had dignitaries from Washington out there to see, to watch it.

I : And this was at the Nevada Chess site?

M : *Yeah, Nevada. They were having a test every week. Practically every engineer that came up with an idea, with a little modify, he'd have it changed a little bit, and then they'd say they'd test it. And they didn't realize for a long time, like I say, every week they were having these bombs go off polluting the air. It took them a number of years before, especially around Nevada there, they realized that it was polluting the air. It was something that was not desirable. And I see that there are some adverse effects on some young kids that had supposedly some problems that were attributed to the tests that were conducted.*

I : Just a few words more about Fermi. On next year will be an anniversary, so do you remember Fermi? Say, some words about him as a person or some personal recollection of Enrico Fermi?

M : *Yeah, he was in our office. He'd sit there and type. He had written the program and actually typed it out himself. Well, you type it like a typewriter and then it'll produce a tape. And then it'll produce this tape that he could load onto the machine. But normally we'd give the program to one of the typists there because they were proficient in typing.*

I : But he would do his own typing?

M : *Yeah, he did his own typing. He sat down and did his own typing and then tried to get it to, people helped him get the tape. And then you just put the tape in the, loaded it.*

I : Was he nice? Was he polite or not so friendly?

M : *Oh, he was very friendly and, like I say, very nice and explained. He lectured a couple of times where he explained stuff to us in simple terms, what was going on and stuff like that. But like I say, he was just interested in doing it himself. Typing it, putting it under, running it through the reader onto the computer. And then asking us to go back to some parts of it, a part of the code or stuff like that.*

I : Do you remember whether he was puzzled or excited by the Fermi-Pasta-Ulam-Tsingou problem? At the beginning, was he excited or disappointed by the results? What did he think of the results of the string problem? Did he care about that?

M : *I don't remember discussing that, about the string problem. That was with Stan Ulam and John Pasta. Those two people were involved in that. Like I say, I remember them saying: "now we've got a computer, we've got to think of problems that could not be solved before, that supposedly these computers could solve."*

I : You said they were surprised with the results?

M : *Yeah, they were surprised.*

I : What did they say about that?

M : *Well, they thought that eventually it would spread out, but it never did. It sort of came back again. The longer we ran it, it confirmed what they didn't think was going to happen. It confirmed what they didn't expect.*

I : Do you remember them saying anything about that?

M : *No, I don't remember much what the situation was.*

I : So about your activities in Los Alamos after those years, which was your work at Los Alamos? You worked in which division?

M : *Yeah, T Division. Radical Division.*

I : And what did you work on? Did you work on the super magnets?

M : *Yes, that magnetic ring, for the fusion, the fusion group, it was one of the groups.*

I : Did you ever feel having problems in your career because of being a woman?

M : *No, they never made an issue of that. They only expect that we were second, that the guys, that the men were superior, supposedly, they got the more interesting problems. And we, women, would get the routine stuff and things like that. The guys were always treated in a higher regard than us. So a lot of us just plain numerical calculations, do columns and columns of additions and then use them to solve the numerical problems. At that time, for some reason, it was that year that I came, and they had recruited a lot of women. So like I say, I think the Korean War was on, and the guys were*

being drafted. So they didn't like training people, take them six months or so to train, to train us. And then the young guys would be drafted. And so that's why they decided to try and recruit women math majors. I was a little surprised. My math teacher said, did you see, I just put it out on the bulletin board. There are some people coming next week from Los Alamos. I didn't know what she meant, Los Alamos. But she says, oh, they're looking for specifically for a math major. And I thought you might like to interview with one of the people that's coming.

A long dialogue of the authors with Ann Menzel concerning important scientific material preserved by M. Tsingou-Menzel follows. The material includes letters and hand written drafts by Von Neumann to Ulam. Here is what Ann Menzel told us.

Okay. Yeah, so like I said, I found these handwritten Xerox copies of notes, and I think they were to Stan Ulam, and I think they were from John von Neumann. There was some... Oh, then I also found this: he was driving back someone to the airport when he wrote some stuff down. Yeah, and it was funny because when I went through the box [containing the quoted material], there's written from probably 10 years ago, very clear directions from my mother: "Annie and Carol, these papers are John von Neumann's, who some consider to be one of the greatest geniuses of our times. He is listed in the encyclopedia. Some of his are originals from Princeton. Don't throw them away. Nothing else."

I : You were good friends with Stan Ulam, right? So, is there something personal about Stan Ulam that you can tell us?

M : *Well, he'd come in for a short while, but would not stay in the office very long. We usually started at 8. He'd come in about 9 or 10 o'clock and stay around for maybe..., a lot of people wanted to talk to him. And he'd be there for a little while. But he'd be gone before noon. He'd come in, like I said, about 10 o'clock and before noon he was gone back to his place.*

I : So, that's typical of mathematicians. They don't stay very much in the office. Was he speaking a good English, Stan Ulam?

M : *Oh yeah, he was fluent in English. I mean, I know no trouble when he spoke.*

I : Was Fermi speaking in English with an Italian accent?

M : *I don't remember.*

I : Your first language is Greek, your second language is Bulgarian, and English is your third language, right? So you had a hard time discerning where an accent comes from. Tsingou is a Greek surname?

M : *Yes, it's a Greek surname. My parents lived in Bulgaria, right over the Greek border. They spoke Greek at home, and they had the Greek schools.*

I : When did they leave?

M : *Papu [her father] left Bulgaria,... I think he came first a couple of times to the US. He was here during the First World War. I mean, during the turn of the century, 1900. In fact, they were recruiting them for the First World War. He became a US citizen, but he says he's from Bulgaria. He's going to go back to the country. He was called out for service, but he claimed that he was involved with the Bulgarian government, and he was going back. And so, he was exempt.*

I : Did he go back to Bulgaria to get you guys, or did he just have you come at World War II?

M : *No, no, he went there and married my mother and came back here.*

I : So, he was in the US first, and then he went back and married?

M : *Yeah.*

I : And then he came back again?

M : *Yeah, he came back here, of course. Kids were born. Well, Sally and I, my sister and I, were born in the US.*

I : But then you went back to Bulgaria again?

M : *Well, that was later on. There was a big depression here, around 1929-30, and the government came to a complete standstill. It was this "russo" that came in and tried to, they thought the American government was going to collapse because no one was producing things. There were no jobs.*

I : So, you went back to Bulgaria during the depression?

M : *Yeah. My parents decided, well, they had some property there and a store, and they thought they'd to go back and live there. But the country*

had come to a complete standstill in this country. People weren't coming up with any new ideas. Everybody thought that there was nothing but going backwards.

I : So, you moved to Bulgaria, you were very young, and then you came back to the US?

M : *It was World War II. I was just six, I think, when we went, and we came back when I was 11. We came back in 1940. In fact, we arrived in New York. We were in Italy for a whole month waiting for a US vessel. But the war was coming, and a lot of these big people that were in the government were sending their families to America, because they knew Greece was going to go to war. Italy had declared war. They were marching down the streets, I remember, in Italy. We were there for a month in Italy, waiting for one of the US vessels. A lot of these top people were putting their families down.*

I : But then you got on one of the last ships?

M : *One of the last ships that sailed from Italy. Well, my mother said she wanted to go on a Greek ship with Greek food and stuff like that. So, we went from Bulgaria to Greece, but then they weren't sailing. We were there in Greece about at least two or three months waiting for it. My mother was notified from America, because Sally and I were American citizens. My mother was a Greek citizen. We were notified that there was an American ship sailing from Italy, Genova. We were able to get on the ship. We crossed the ocean and were there within a week or something. But we waited there for a long time.*

I : And then where did you land in the US? In New York?

M : *In New York, yeah. That was December 1940. And then Pearl Harbor came that December 1940. And then they moved. They went to Milwaukee, Wisconsin.*

I : Did your father have a popcorn truck or he had a business in Wisconsin?

M : *Mostly he had restaurants.*

I : Okay, so he worked at a restaurant in Milwaukee. ~~And~~ That's where you grew up?

M : *Yeah. I went to college at the University of Wisconsin in Milwaukee. First in Milwaukee, then in Madison. Well, see, first we started school in*

Milwaukee. I didn't know English when I came. I went to school and I learned English because Sally, my sister, was a year ahead. So she went to school. She didn't know my English. So they kept her an extra... I think they put her in the third grade and me in the second grade.

Here is another anecdote by Ann Manzel.

So my aunt, my mom's sister, who's also still alive, she's 97, her name is Sally Peavy. Single was her maiden name. She worked on the Johnniac at the Bureau of Standards in Washington. At the same time, during the same era. Many, many years. Yeah, and it was very funny because when my children were small, we were visiting the Smithsonian with them and there was a display on both the Maniac and the Johnniac and the first computers. This must have been 20 years ago, my mom and my aunt started to have a sibling rivalry debate over which computer worked better and who did what. And people thought that they were docents and this whole group was gathering around as they were going back and forth over who did what, when, with these first computers. It was really funny. And my kids who were small, five or six, we were looking for them and they came running and they're like "mommy, mommy, aunt Sally and grandma are having a big argument and people are watching!"

I : Weren't you, for the lab, the person who worked on Affirmative Action for the Los Alamos lab?

M : *Yeah, to check that they hire women. That they're given the same pay. I was supposed to ask them how come they're not hiring this woman who had the same qualifications or even better. But the guys weren't too happy working with women. So they preferred to recruit. But then the regulations came in that they had to get a certain percentage of new recruits had to be women and minorities, people from a lot of different groups rather than U.S. citizens, some foreigners as well that they tried to recruit.*

And anyway, I was there for, I can't believe, for 40 years I worked at the lab... I came in '52 and I retired in '92. So 40 years later, exactly 40 years later, I thought I was planning on working. I wasn't planning on retiring. I was planning on working another five years. But then they came along and

offered us an incentive. They gave us extra. They added five years to our retirement. Otherwise, if it wasn't for that, I would have worked because I enjoyed my work and found it interesting. So even though my husband had already retired by that time, I was still interested in continuing working a few more years. But like they say, they offered us this extra five years towards our retirement. Then I accepted it and stopped working.

I : We thank you very much. Mary, it was an honour.

M : *Okay. Thank you. Have a good day. Bye-bye.*